DAC (Digital to Analog Converter)

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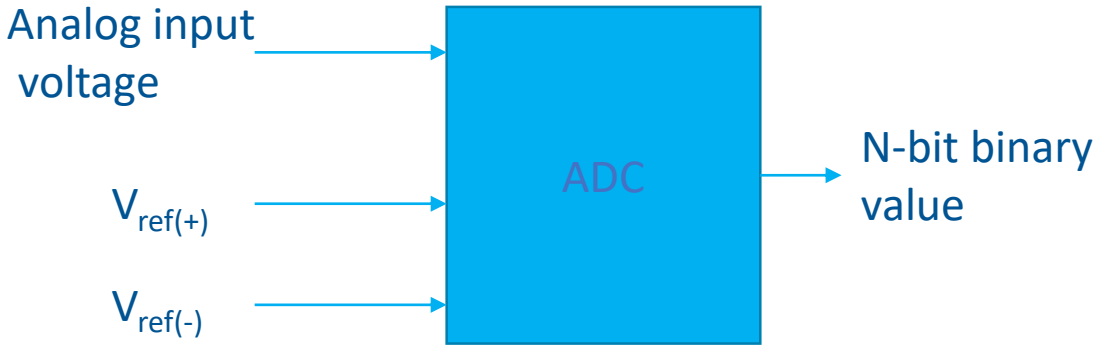
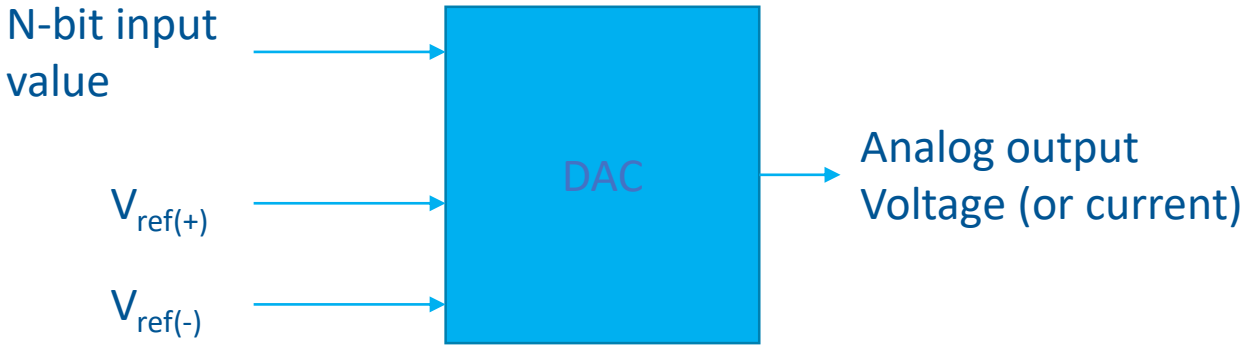
https://www.electronics-lab.com/article/digital-to-analogue-converter-r-2r/

ADC


Analog input voltage
$V_{ref(+)}$
$V_{ref(-)}$
ADC
N-bit binary value

DAC


N-bit input value
$V_{ref(+)}$
$V_{ref(-)}$
DAC
Analog output Voltage (or current)

DAC

It has n input lines and one analog output line.

Resolution is the smallest analog value that can be represented by the n-bit binary code.

Resolution = $V_{ref}/2^n$ volts

The number of analog steps is 2^n-1.

Every analog output is the multiple of resolution.

Example

Example: if n=4 (4-bit ADC), Vref = 5V, then

Output voltage = $D_{input} * V_{ref}/2^n$

DigitalInput	DecimalValue	AnalogOutput (V)
0000	0	0
0001	1	0.3125
0010	2	0.625
0011	3	0.9375
0100	4	1.25
.....	.....	
1100	12	3.75
1101	13	4.0625
1110	14	4.375
1111	15	4.6875

Example

A DAC has full-scale analog output of 10V, and accepts 6 bits input. Find the voltage corresponding to each analog step?

Answer: Resolution = Full-scale output/Number of steps

$$= 10V/(2^6-1) = 0.15873V$$

Example

Find the binary bit size of a DAC if the full-scale output is 12.6V and resolution is 200mV.

Answer: Resolution = Full-scale output/Number of steps

$$200\text{mV} = 12.6/(2^n-1)$$

From this, we find n to be 6.

Example

A 5-bit DAC produces an output voltage of 0.1V for a digital input of 00001. What is the full-scale output? Find the output for an input of 10101.

Answer: Since the output for minimum digital value (00001) is 0.1V, the resolution or step-size is 0.1V.

FullScaleOutput = StepSize*NumberOfSteps

$$= 0.1*(2^5-1) = 3.1V$$

Since $10101_2 = 21_{10,}$ analog output is 21* 0.1 = 2.1V